



FABRIC MANUFACTURING II

Part: B

Course Code: FE 351-0723

Warp Knitted Structures

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Lecturer

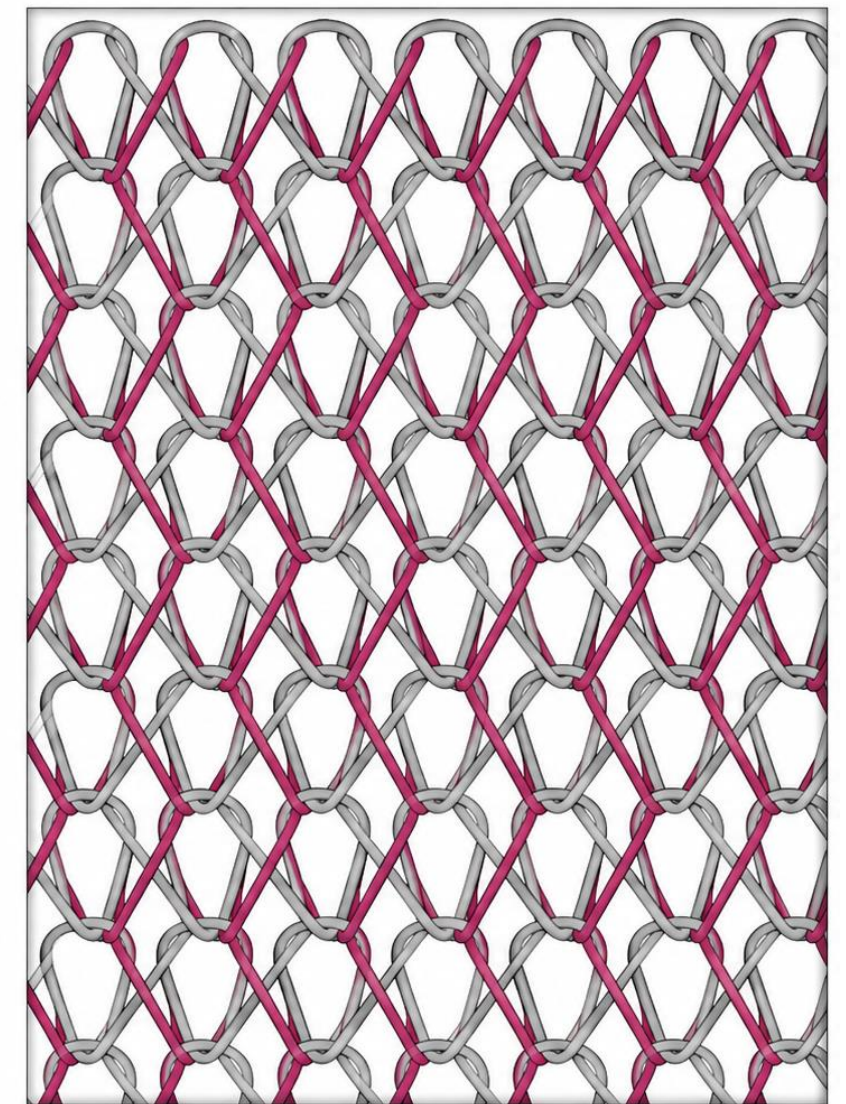
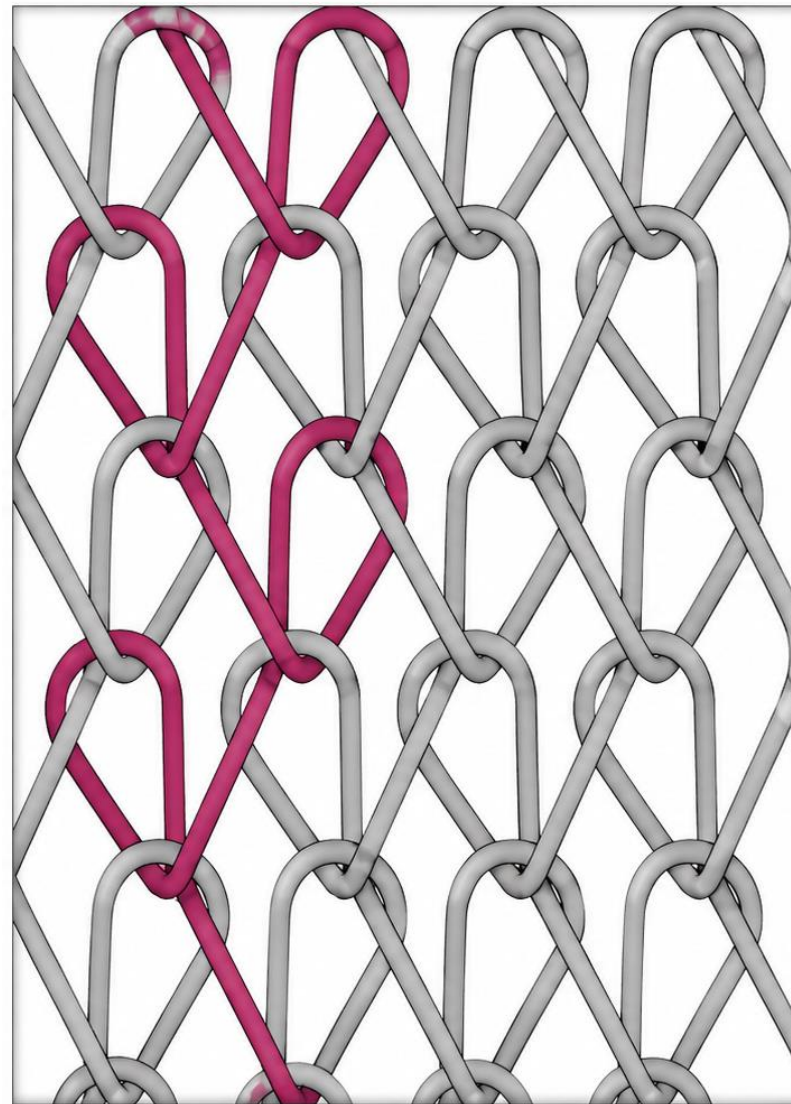
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CLASSIFICATION

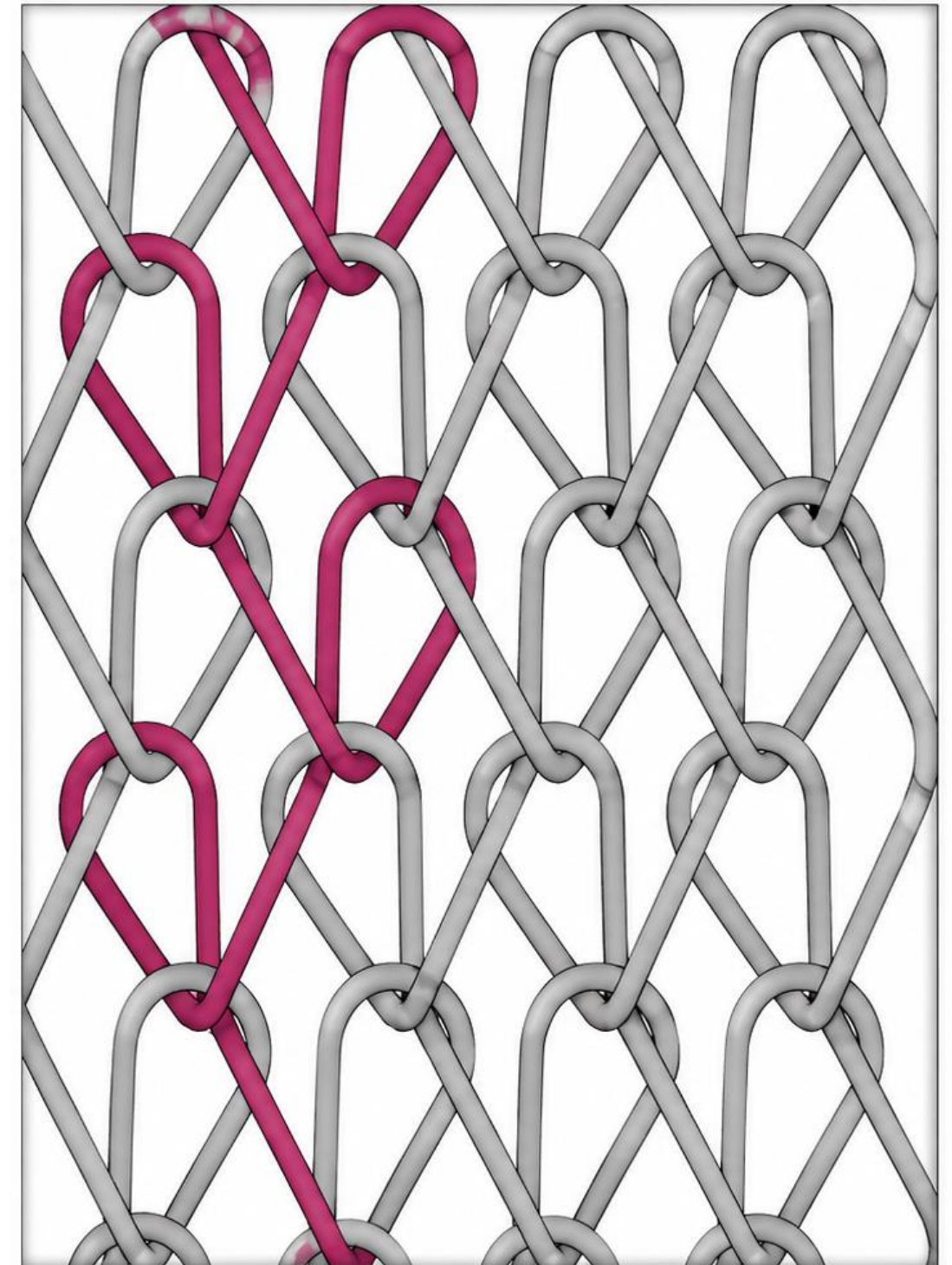
1. Single Guide Bar Structure
2. Double Guide Bar Structure



SINGLE GUIDE BAR STRUCTURE

Characteristics of Single Guide Bar Structures

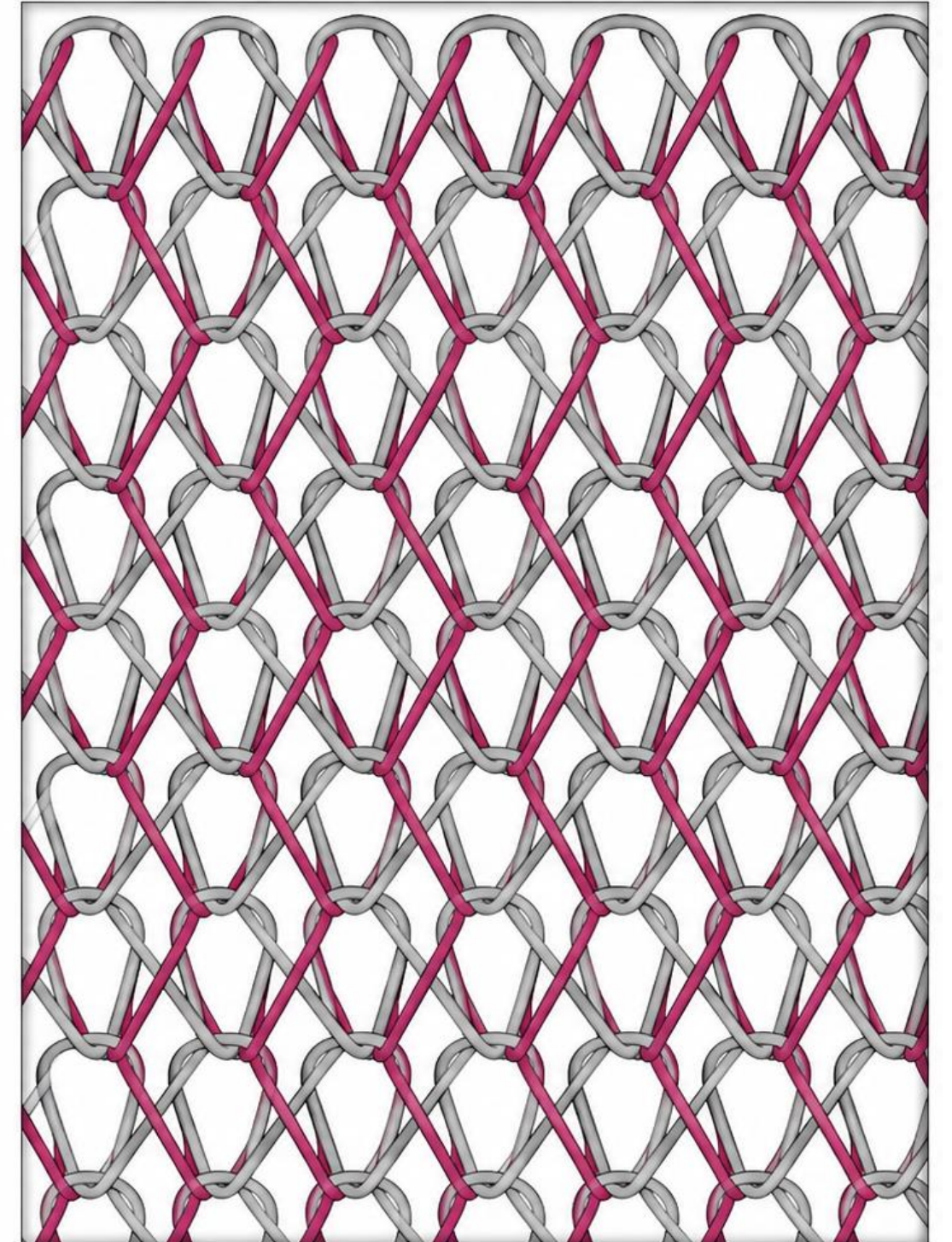
- Usually very unstable (dimensionally).
- Could split easily.
- Unbalanced stitch structure.
- Poor stitch inclination.
- Not suitable for commercial applications.



DOUBLE GUIDE BAR STRUCTURE

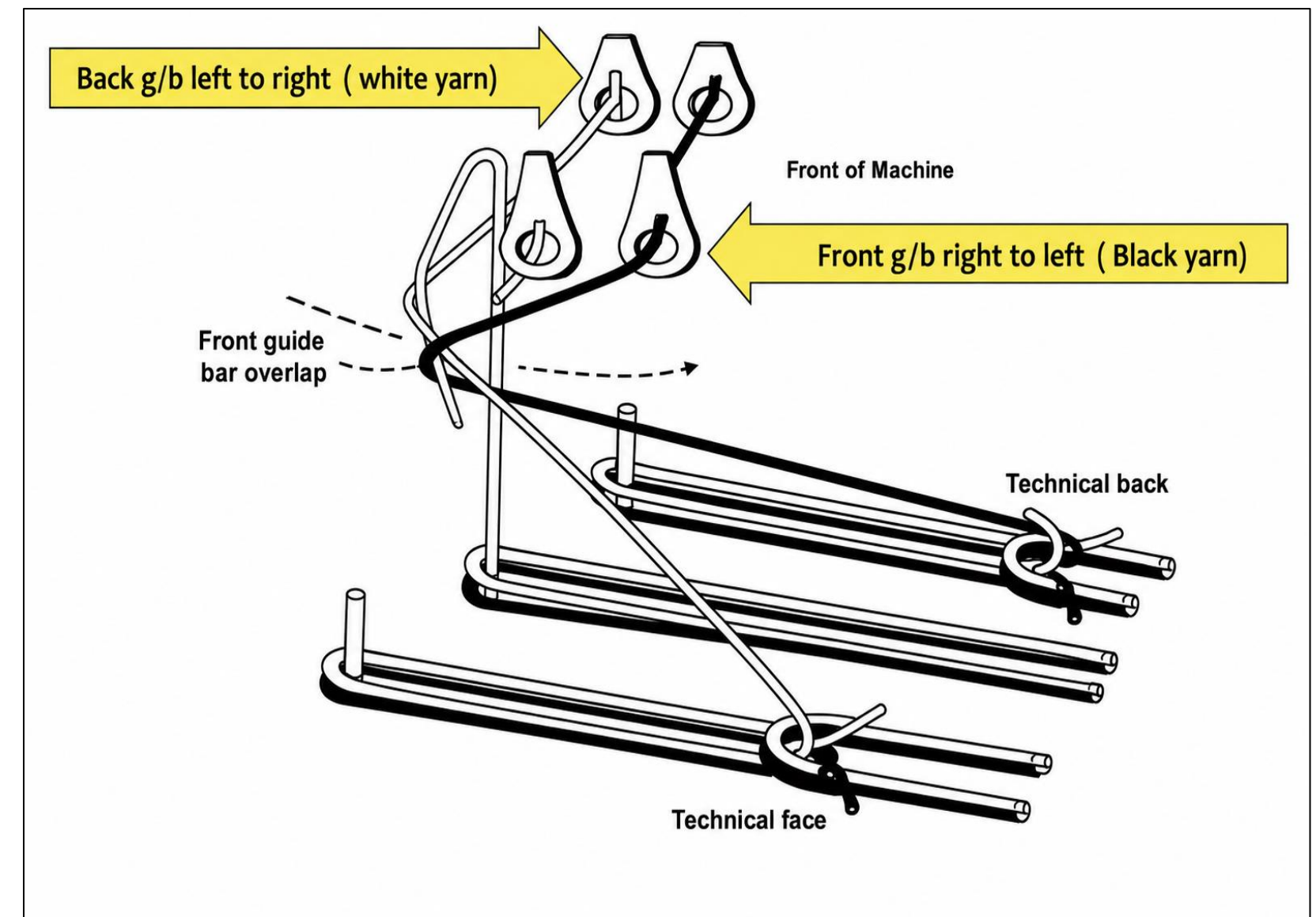
Characteristics of Double Guide Bar Structures

- Knitted with two sets of yarns.
- Each needle is delivered with two yarns.
- Underlaps of the two yarn systems are created in opposite directions.



DEFINITIONS

- ❖ Needle hook side is the front side of the needle.
- ❖ Back side of needles are on front side of the knitting machine.
- ❖ Guide bar furthest away from the needle hook is the Front Guide Bar.
- ❖ Guide bar closest to needle is the Back Guide Bar.



RULES FOR PLOTTING LAPPING DIAGRAM

- The lapping movement of each guide bar must be represented separately.
- The lapping diagrams of all guide bars must be plotted from the same course in order to illustrate the relative lateral position of guides during knitting.

Step-by-Step Guide for Creating Lapping Diagrams

Step 1: Plot lapping movement of first guide bar.

Step 2: Plot lapping movement of second guide bar, starting from the same course.

Step 3: Number spaces between needles from right to left. Number spaces separately for each guide bar.

Step 4: Consider only overlap to generate the chain-links.

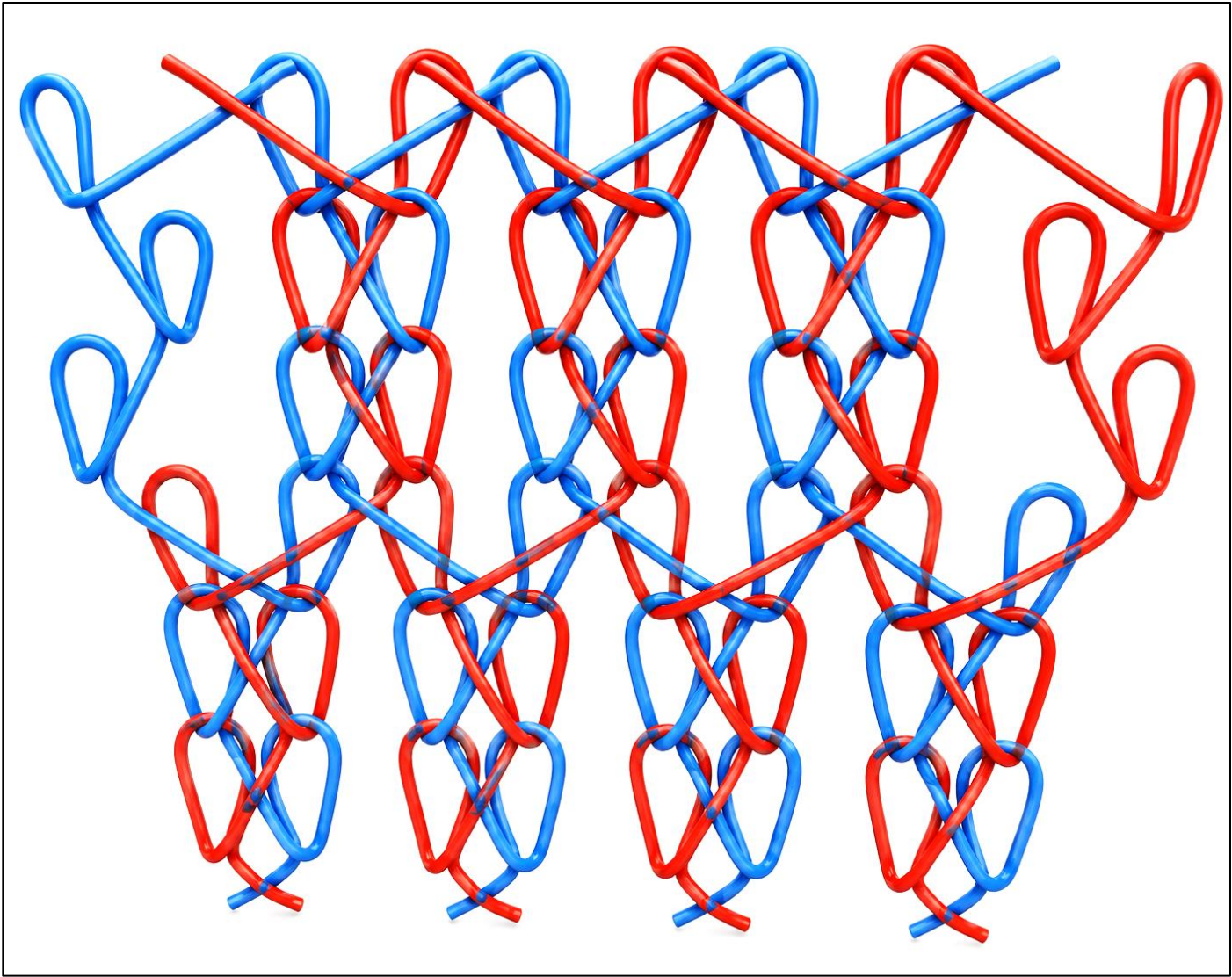
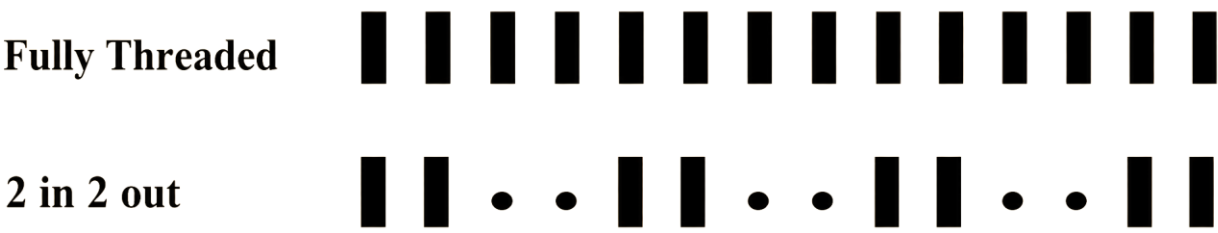
THREADING PLAN OF GUIDE BAR

Defines the threading of yarn through individual yarn guides. Rules for Drawing Threading Plan:

- A short vertical line represents a guide with a yarn.
- A dot represents an empty guide.
- A row consisting of the below symbols represents a guide bar.

Examples

Fully Threaded-



DOUBLE TRICOT STRUCTURE

Threading plan

Front & Back guide bars fully threaded.

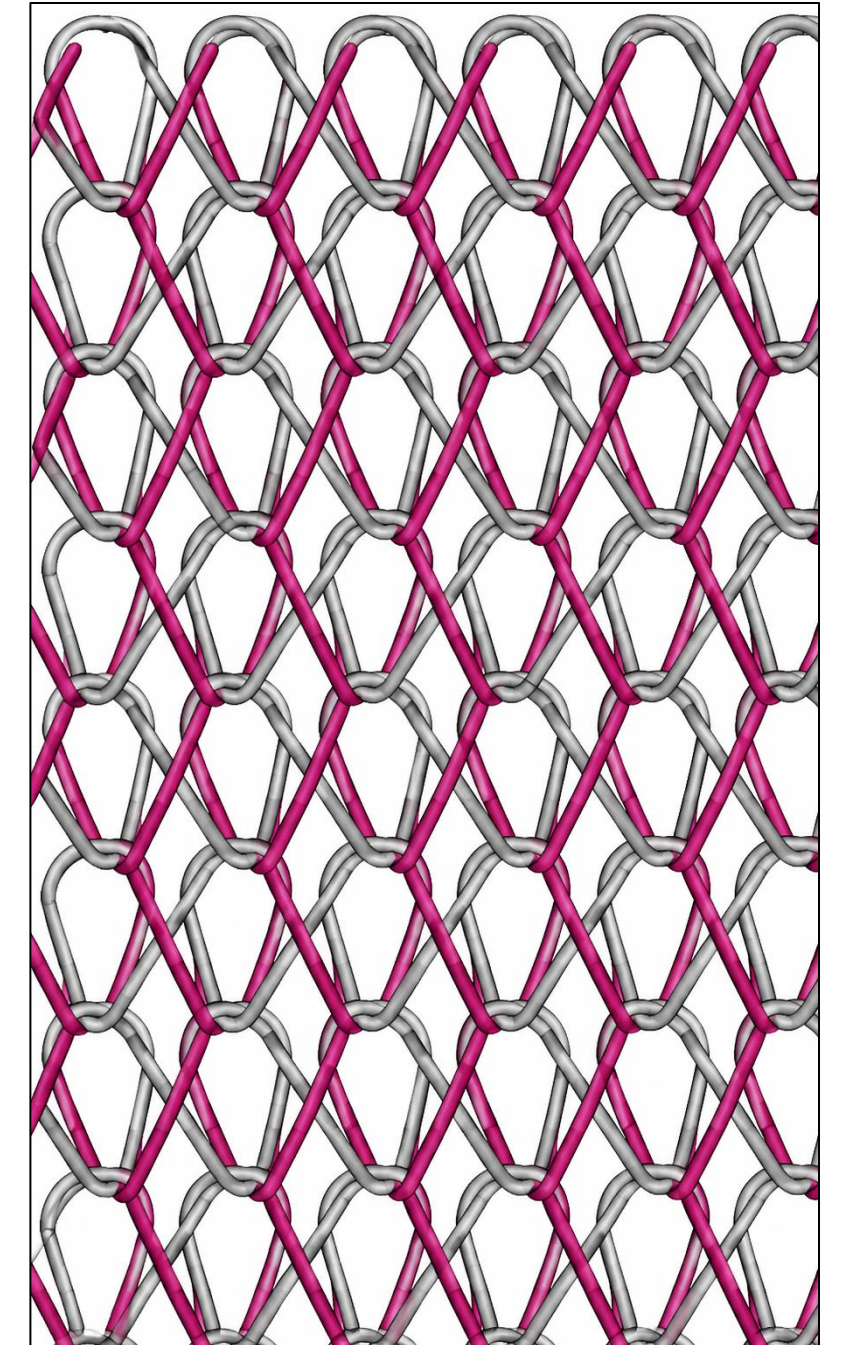
Lapping movement

Front Guide Bar: *1 and 1 lap (Tricot Stitch)*.

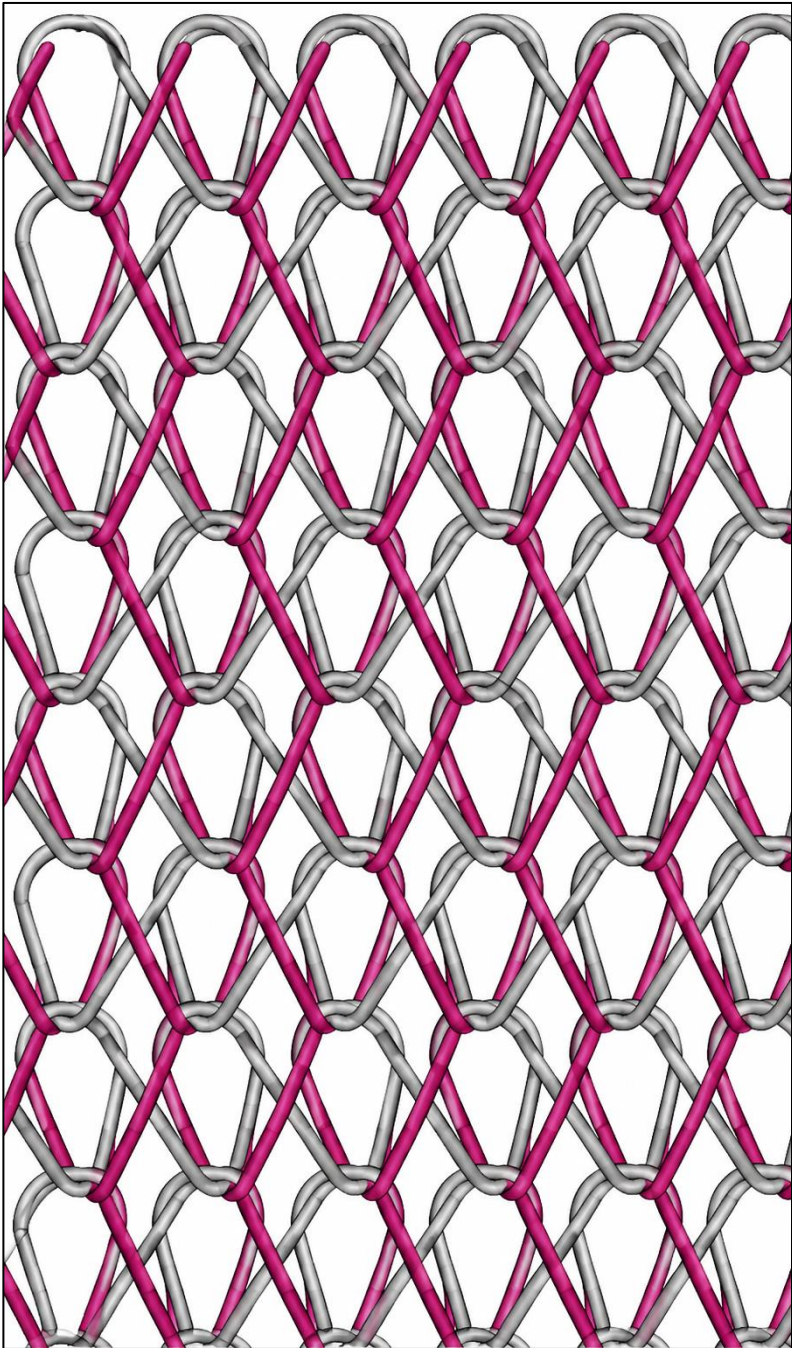
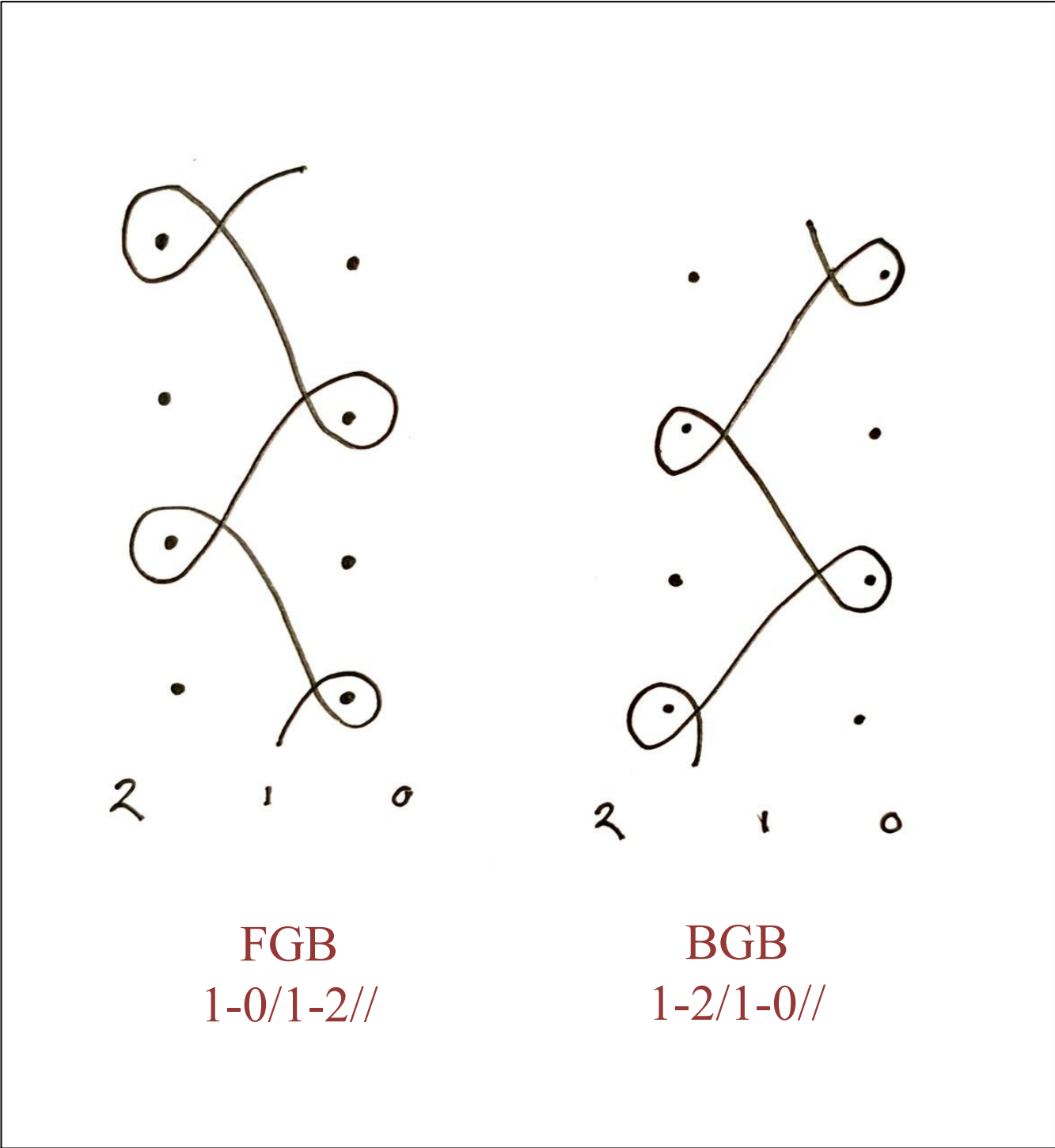
Back Guide Bar: *1 and 1 lap (Tricot Stitch) in opposite direction*.

Fabric characteristics

- Light weight fabric.
- Splits very easily if a yarn breaks or a stitch drops.



DOUBLE TRICOT STRUCTURE



LOCKNIT STRUCTURE

Threading plan

Front & Back guide bars fully threaded.

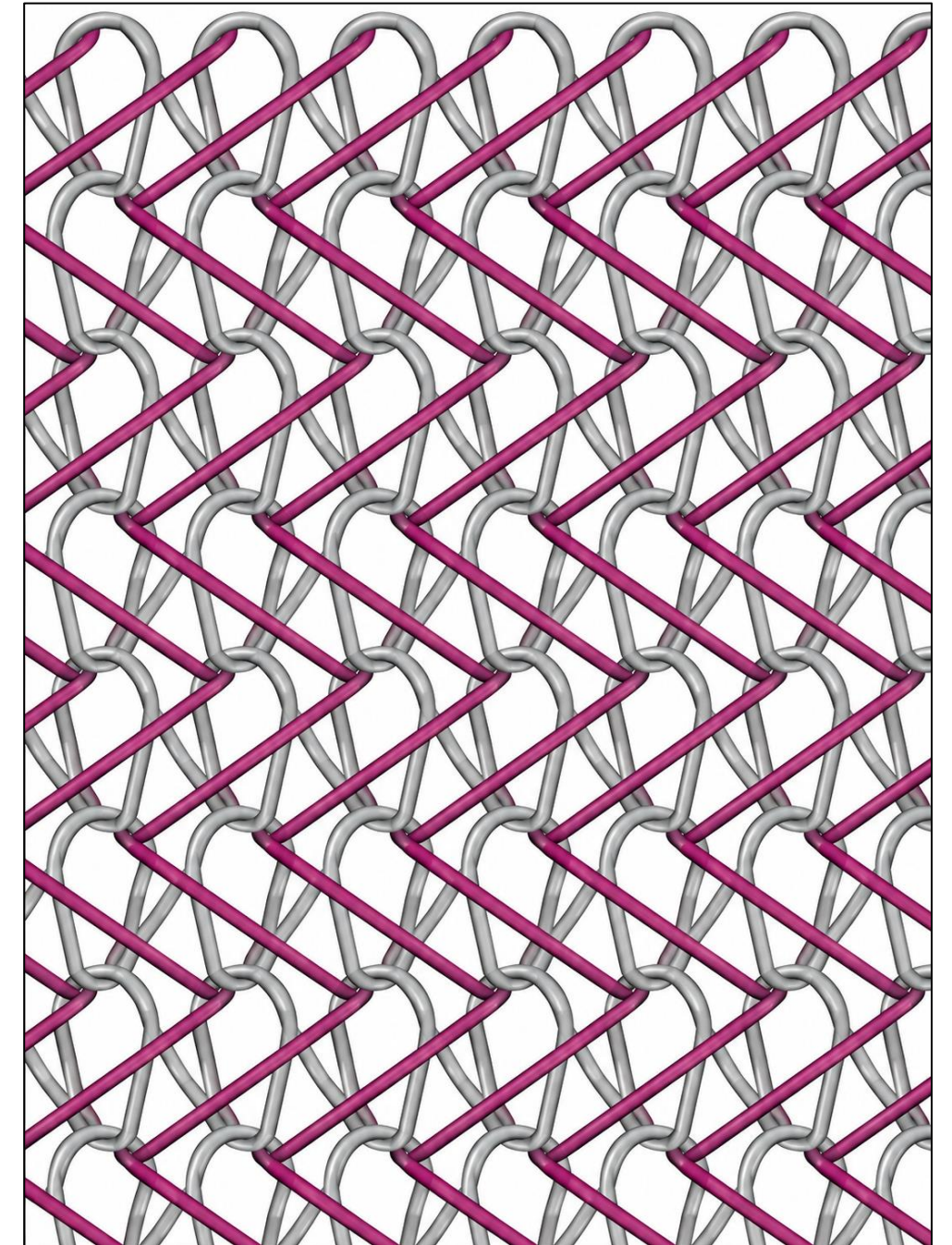
Lapping movement

Front Guide Bar: *2 and 1 lap (Cord Stitch).*

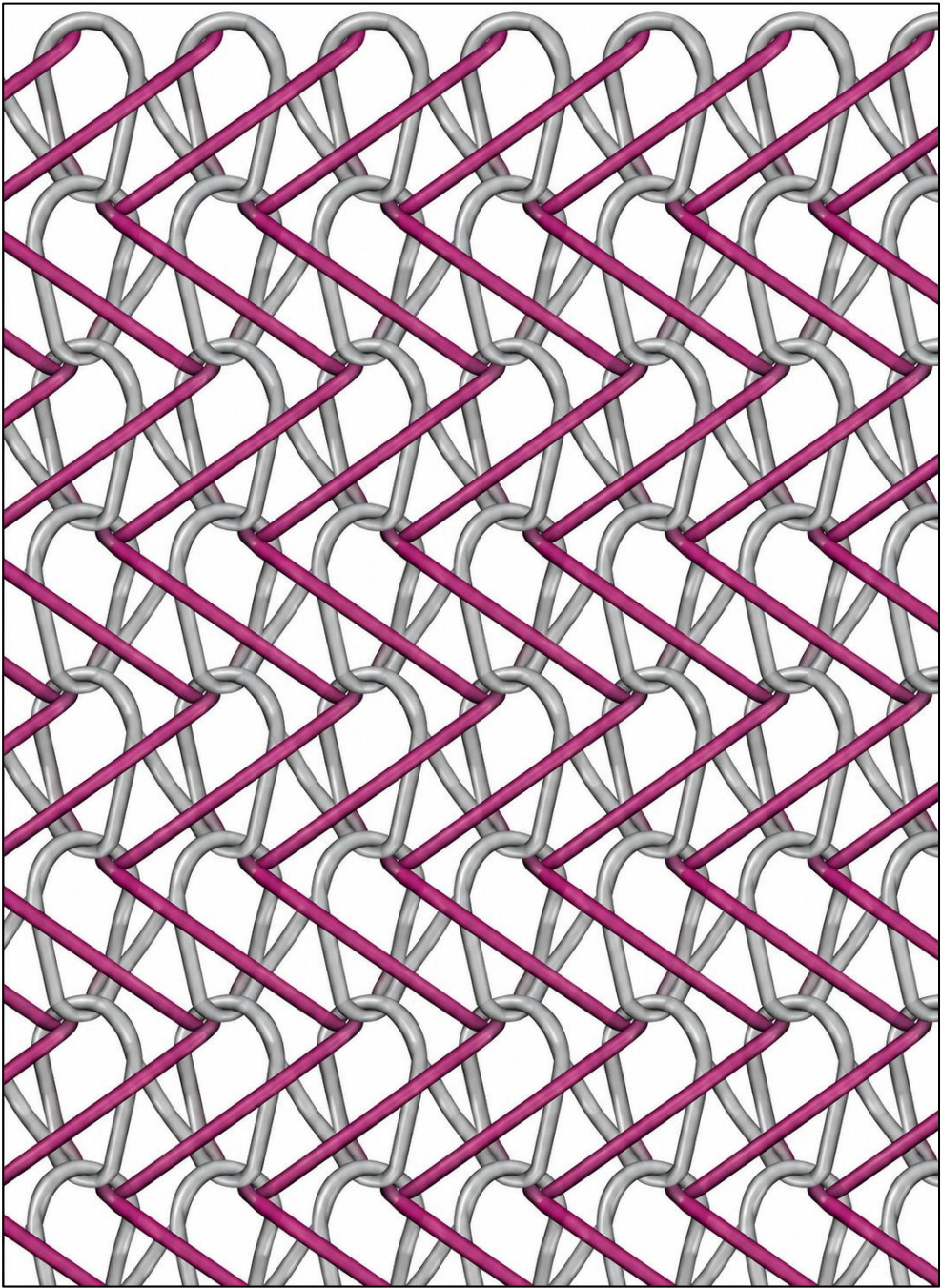
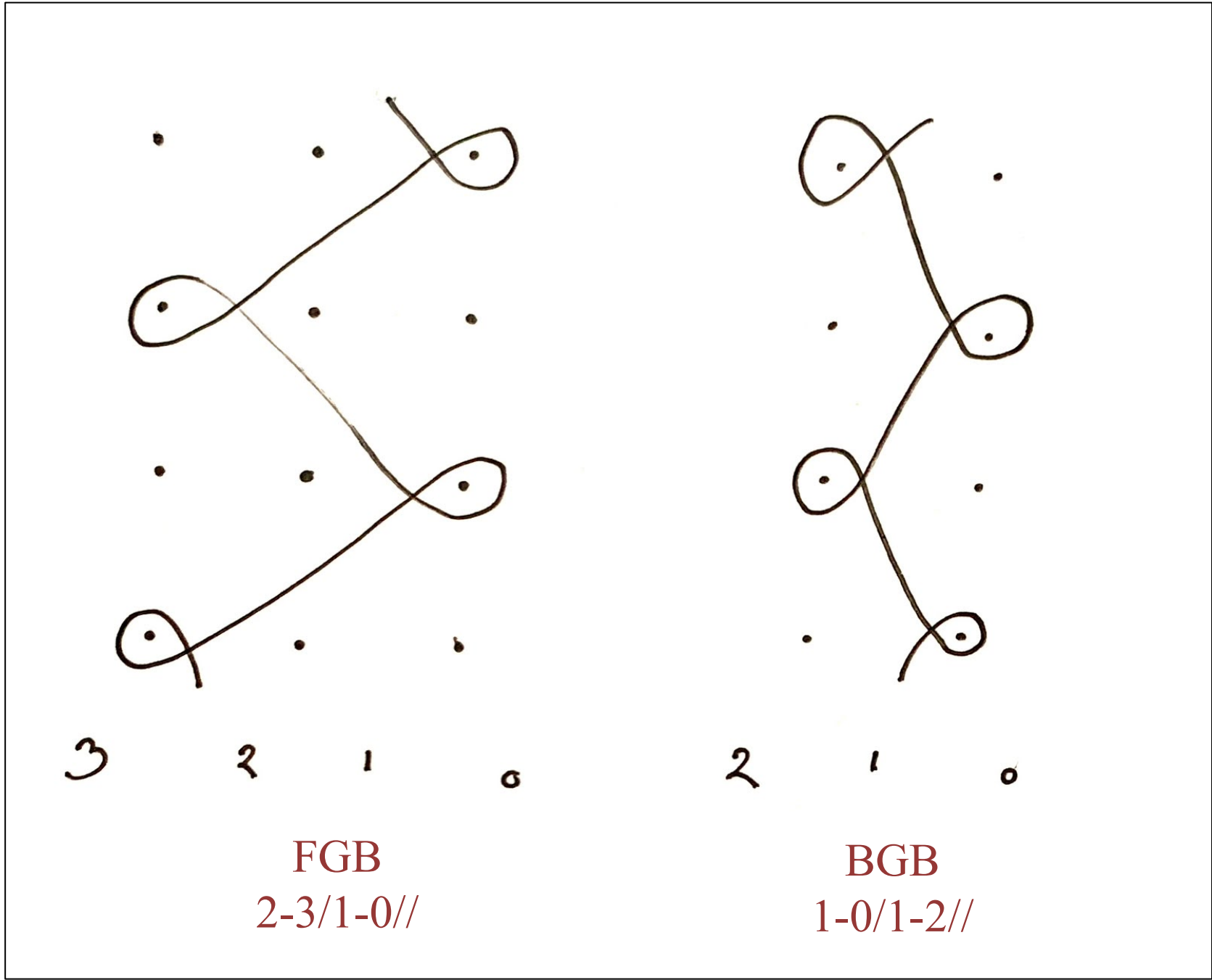
Back Guide Bar: *1 and 1 lap (Tricot Stitch) in opposite direction.*

Fabric characteristics

- Light weight fabric.
- Non splitting fabric.
- Good stretch properties.
- Smooth back due to free floating underlaps.
- Edge curling towards technical back.
- Lateral contraction after knitting by 67%.



LOCKNIT STRUCTURE



REVERSE LOCKNIT STRUCTURE

Threading plan

Front & Back guide bars fully threaded.

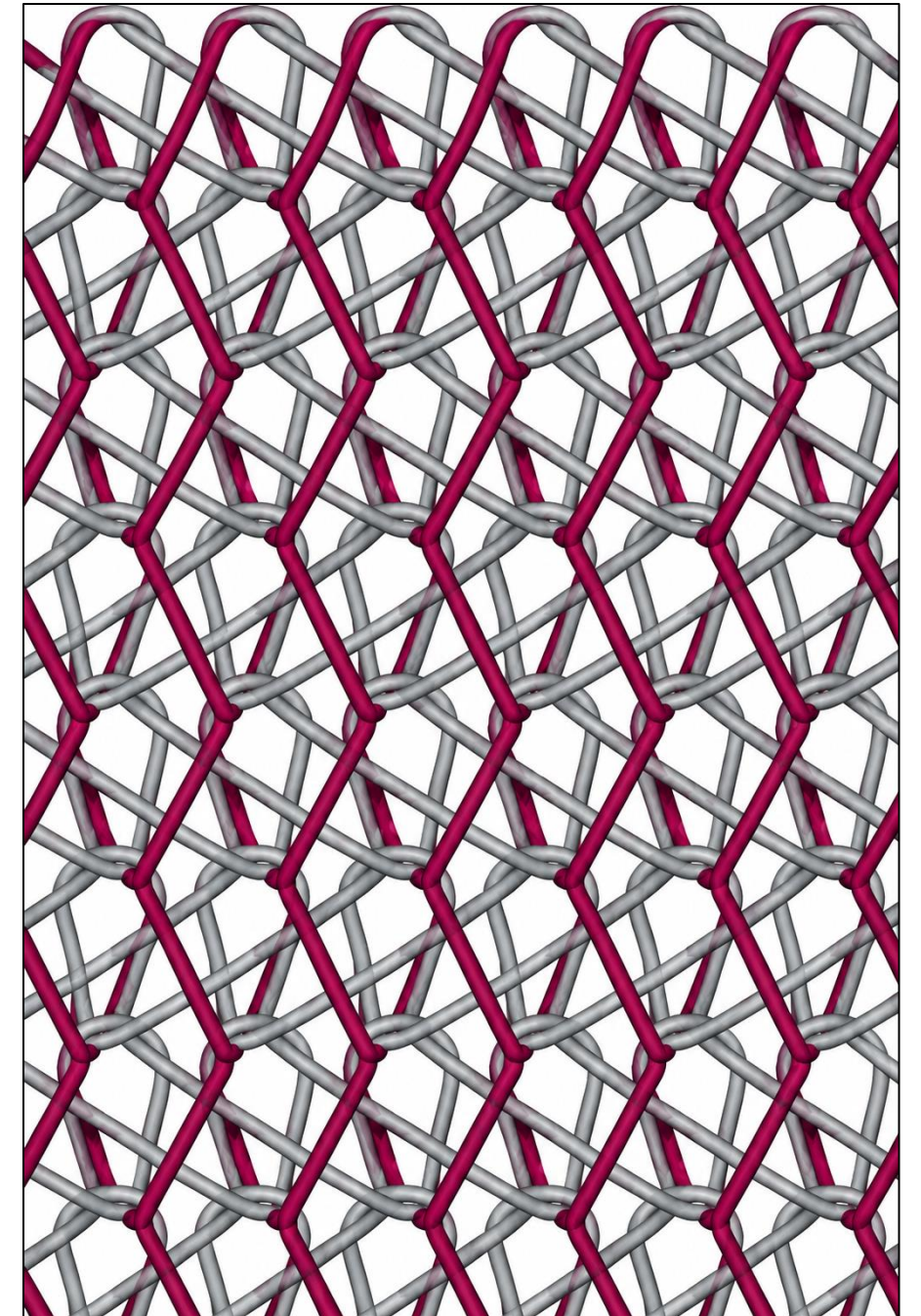
Lapping movement

Front Guide Bar: *1 and 1 lap (Tricot Stitch).*

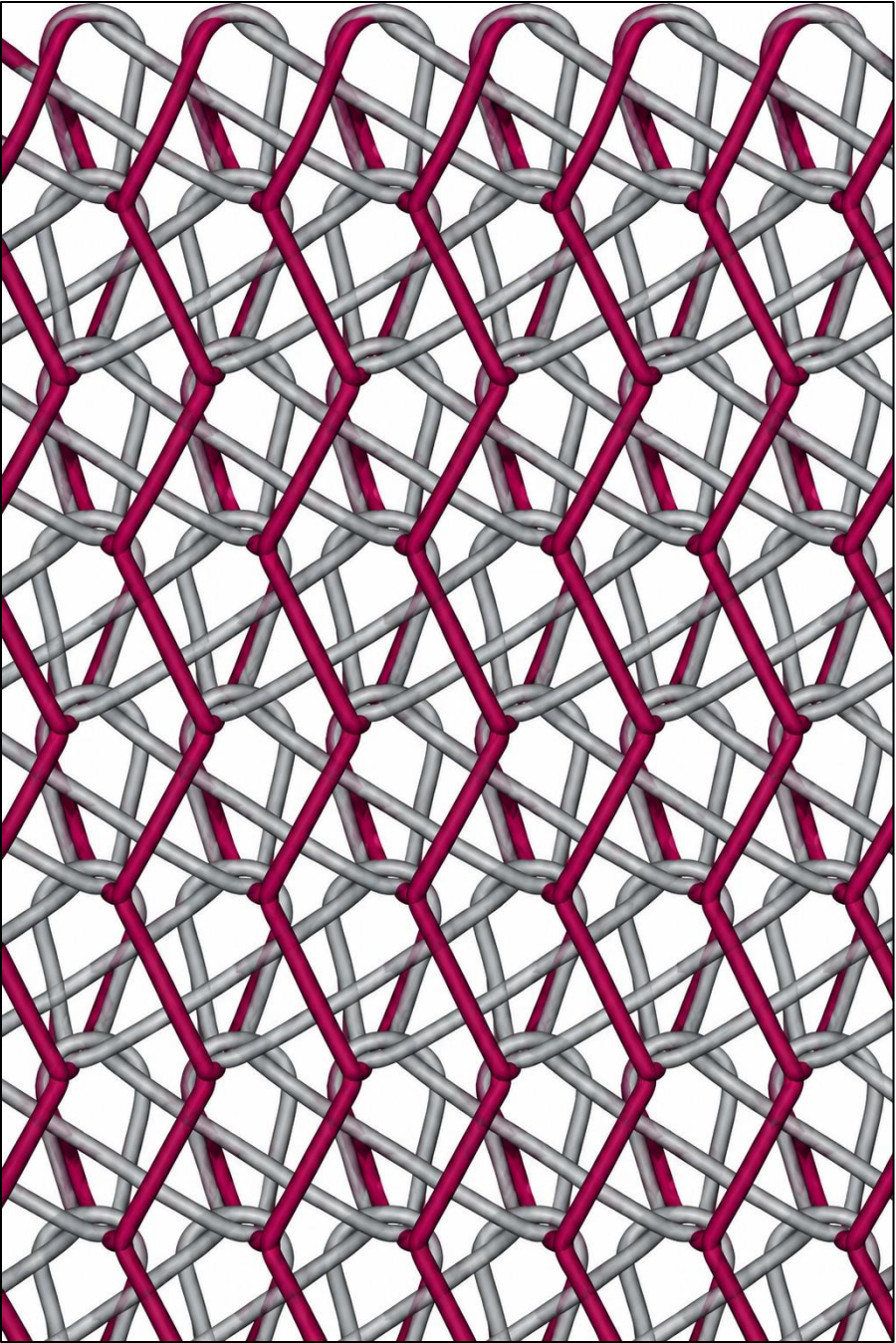
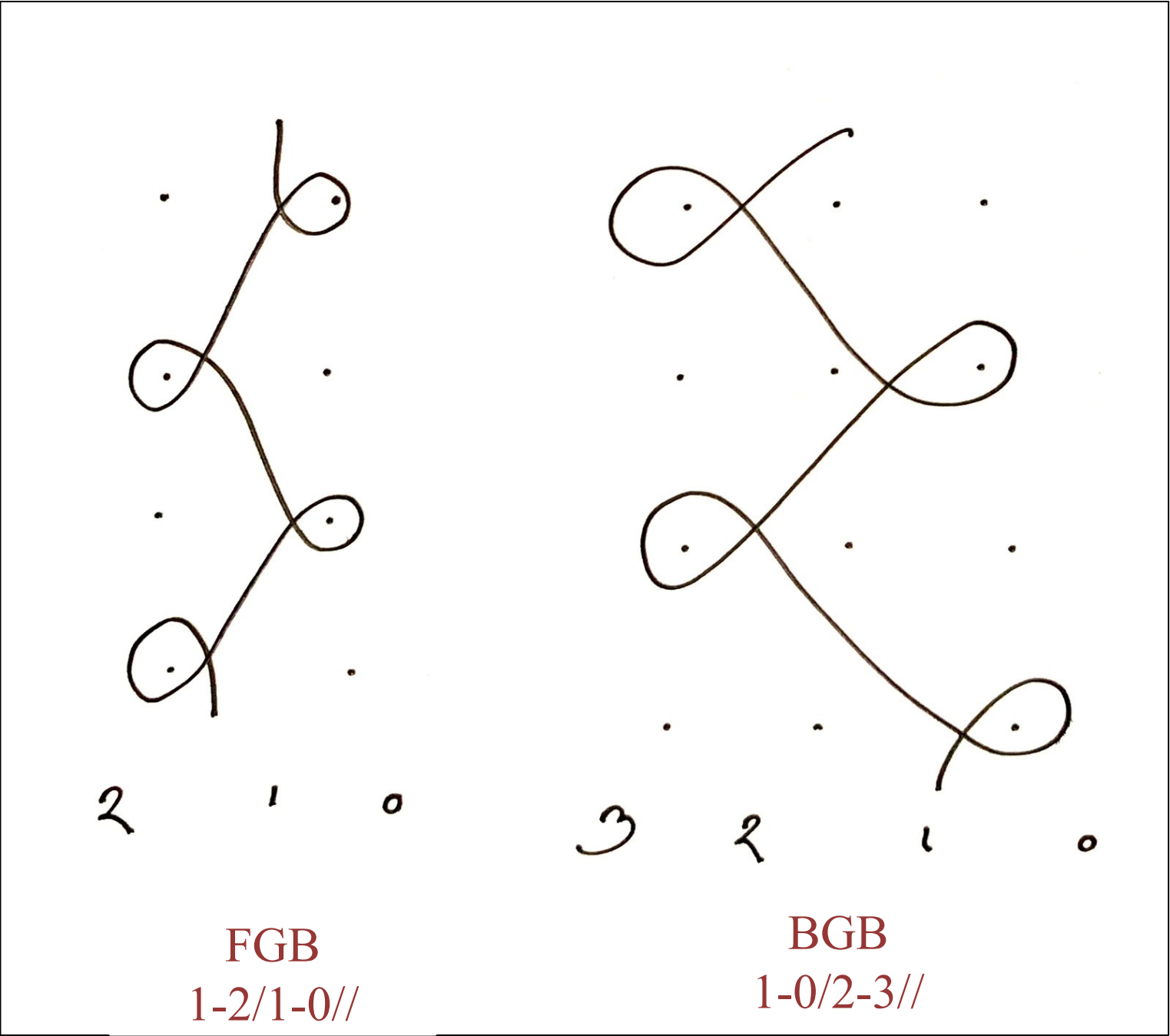
Back Guide Bar: *2 and 1 lap (Cord Stitch) in opposite direction.*

Fabric characteristics

- Not as stable as Locknit fabric.
- Semi stable and stable structures.
- Shrinks considerably on leaving the knitting zone.



REVERSE LOCKNIT STRUCTURE



SATIN STRUCTURE

Threading plan

Front & Back guide bars fully threaded.

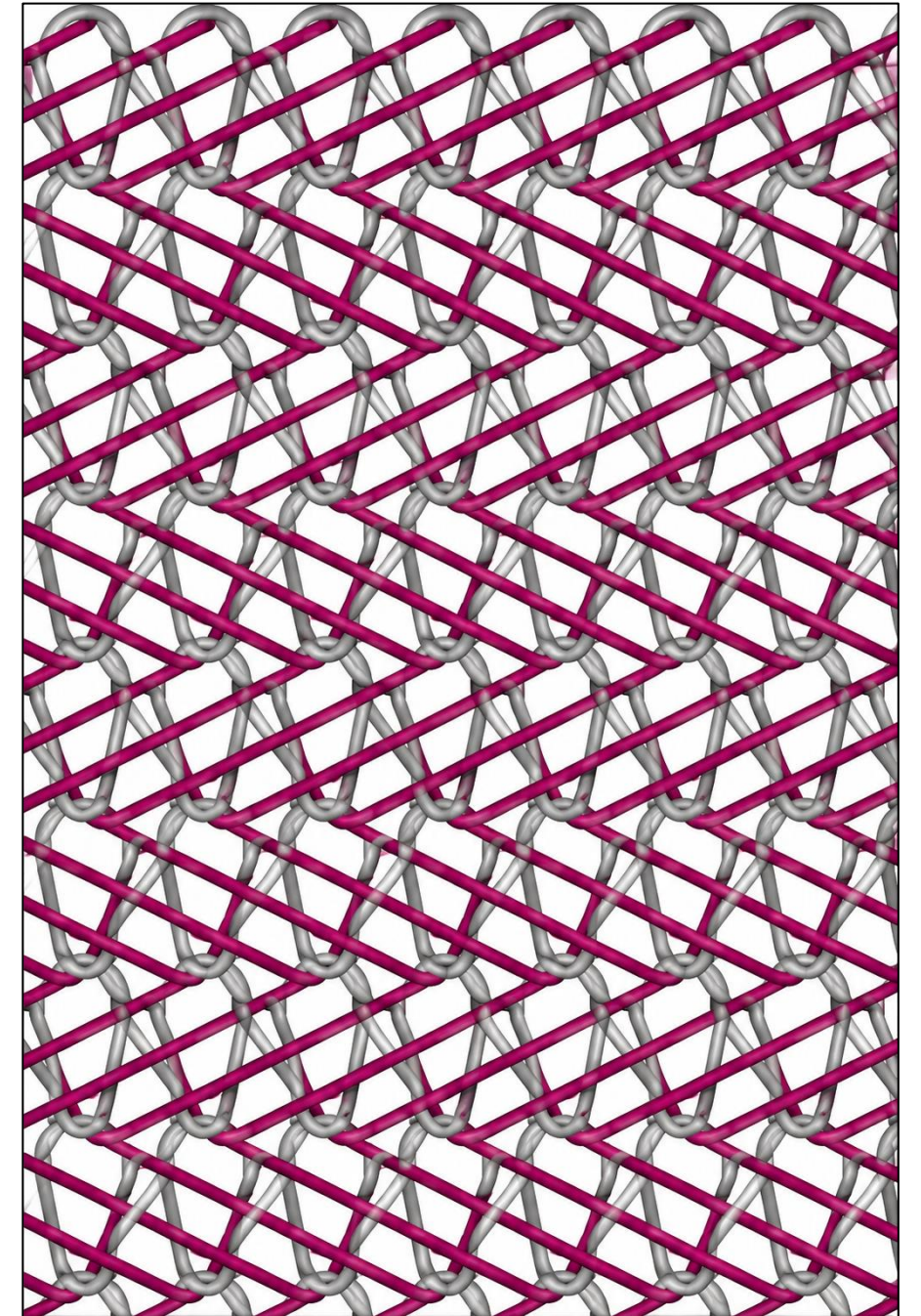
Lapping movement

Front Guide Bar: *3 and 1 lap (3 Needle Structure) or 4 and 1 lap (4 Needle Structure).*

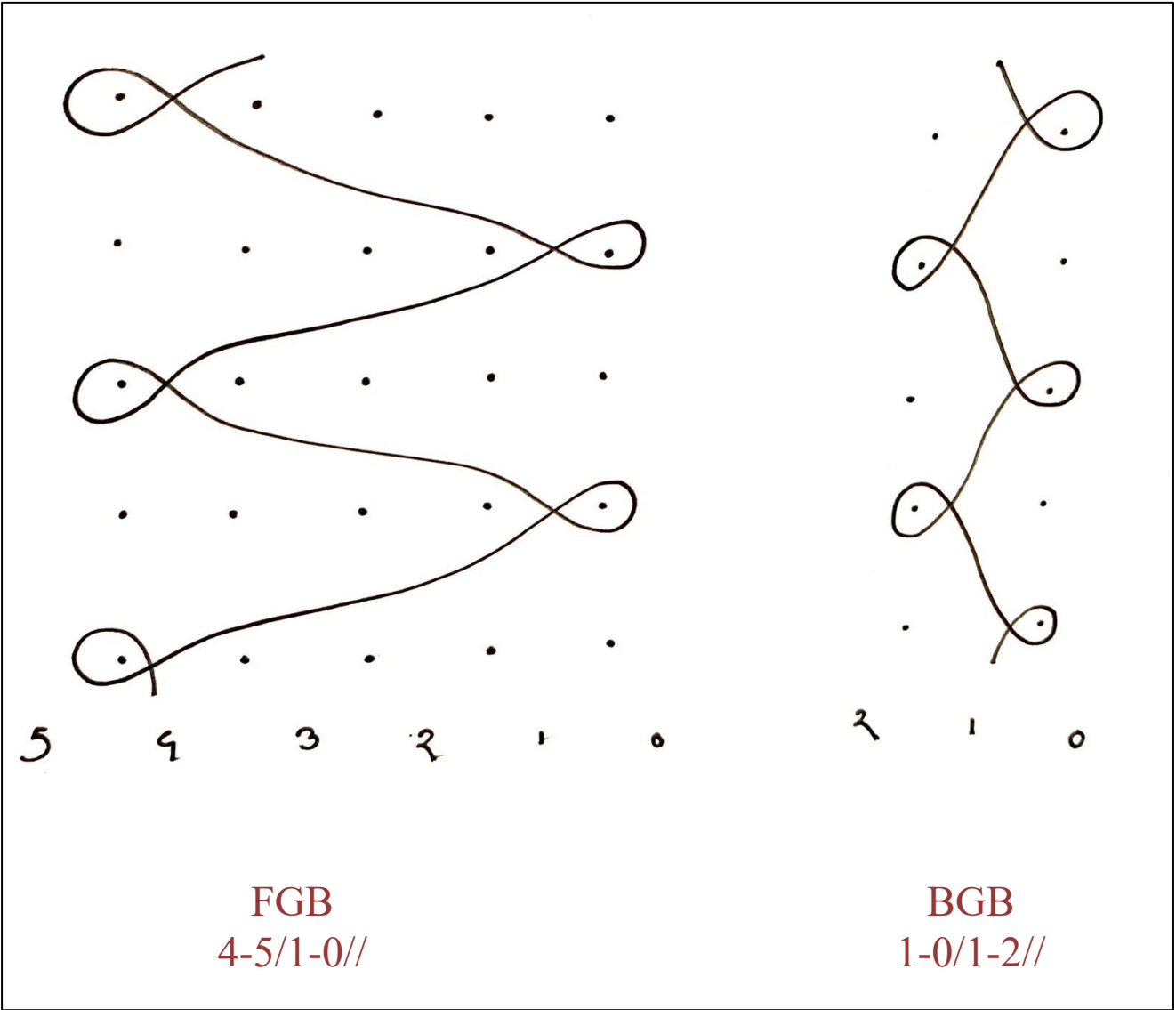
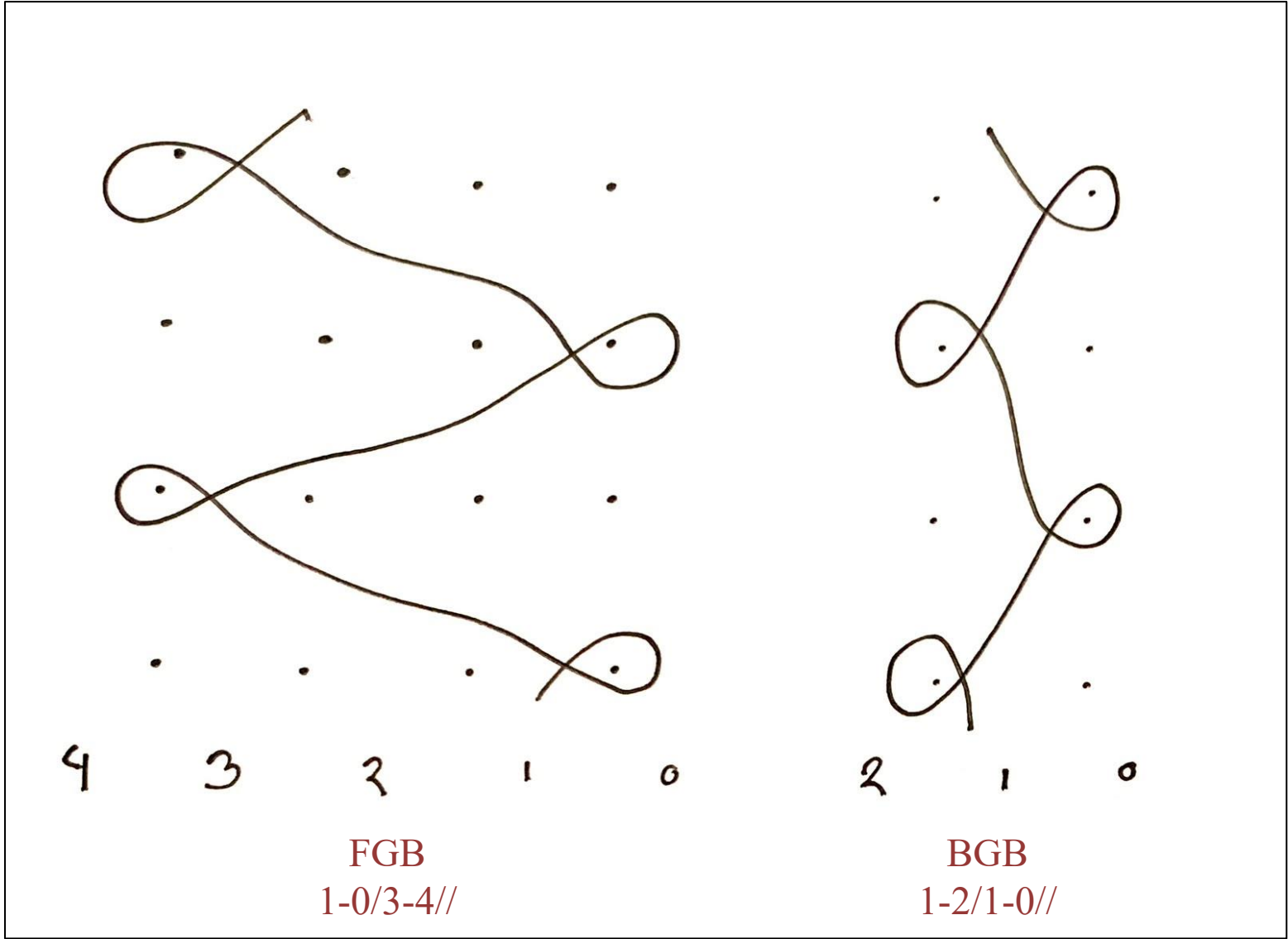
Back Guide Bar: *1 and 1 lap (Tricot Stitch) in opposite direction.*

Fabric characteristics

- Non splitting fabrics.
- Good stretch properties.
- Very smooth back due to free floating underlaps.
- Edge curling towards technical back.
- Lateral contraction after knitting.
- Garments very comfortable to wear.
- Heavier & Elastic.



SATIN STRUCTURE



SHARKSKIN STRUCTURE

Threading plan

Front & Back guide bars fully threaded.

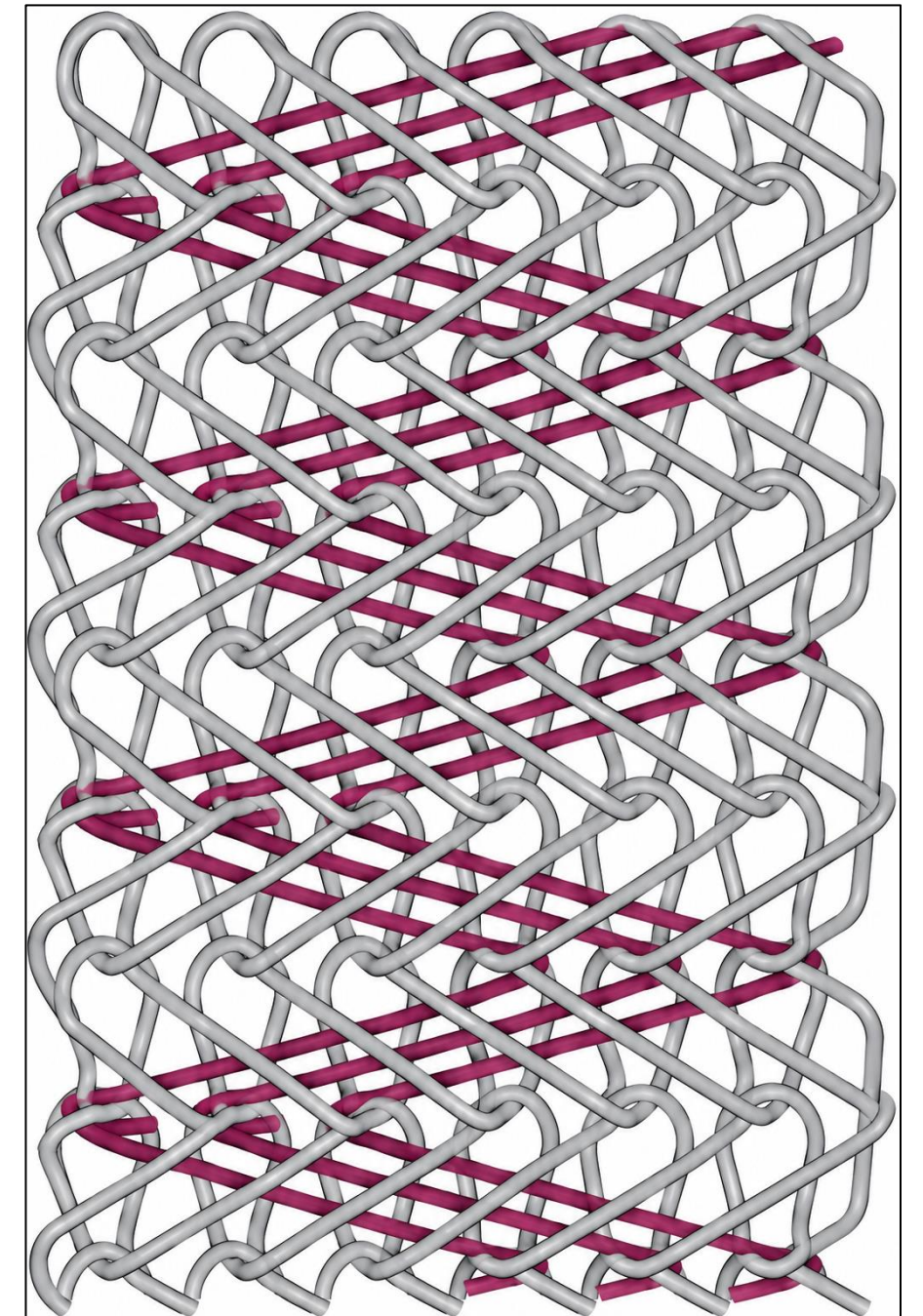
Lapping movement

Front Guide Bar: *1 and 1 lap (Tricot Stitch).*

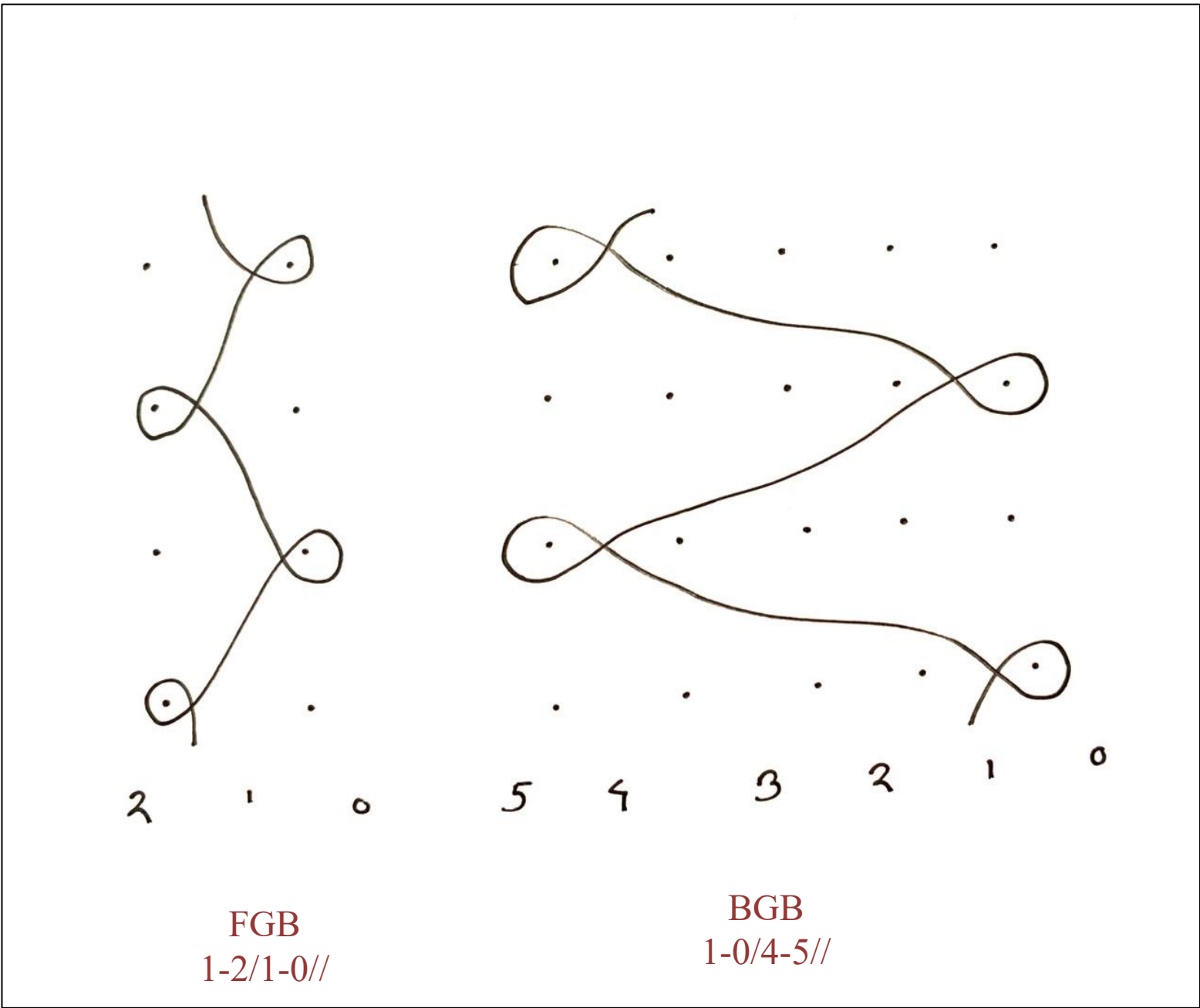
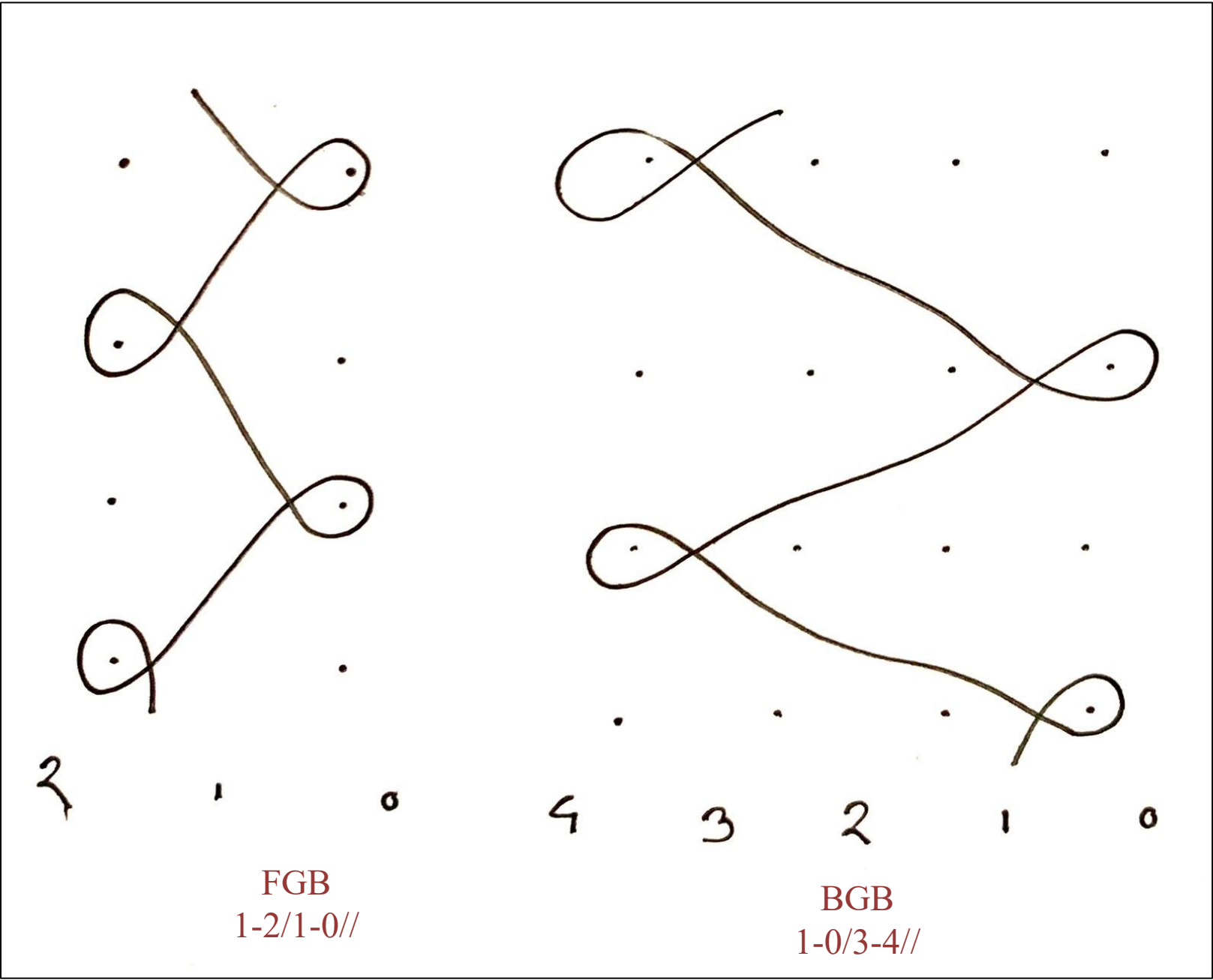
Back Guide Bar: *3 and 1 lap (3 Needle Structure) or 4 and 1 lap (4 Needle Structure).*

Fabric characteristics

- Trapped longer underlaps restricted fabric shrinkage.
- Rigid and very stable structures most stable.
- Rough technical back.



SHARKSKIN STRUCTURE



QUEEN'S CORD STRUCTURE

Threading plan

Front & Back guide bars fully threaded.

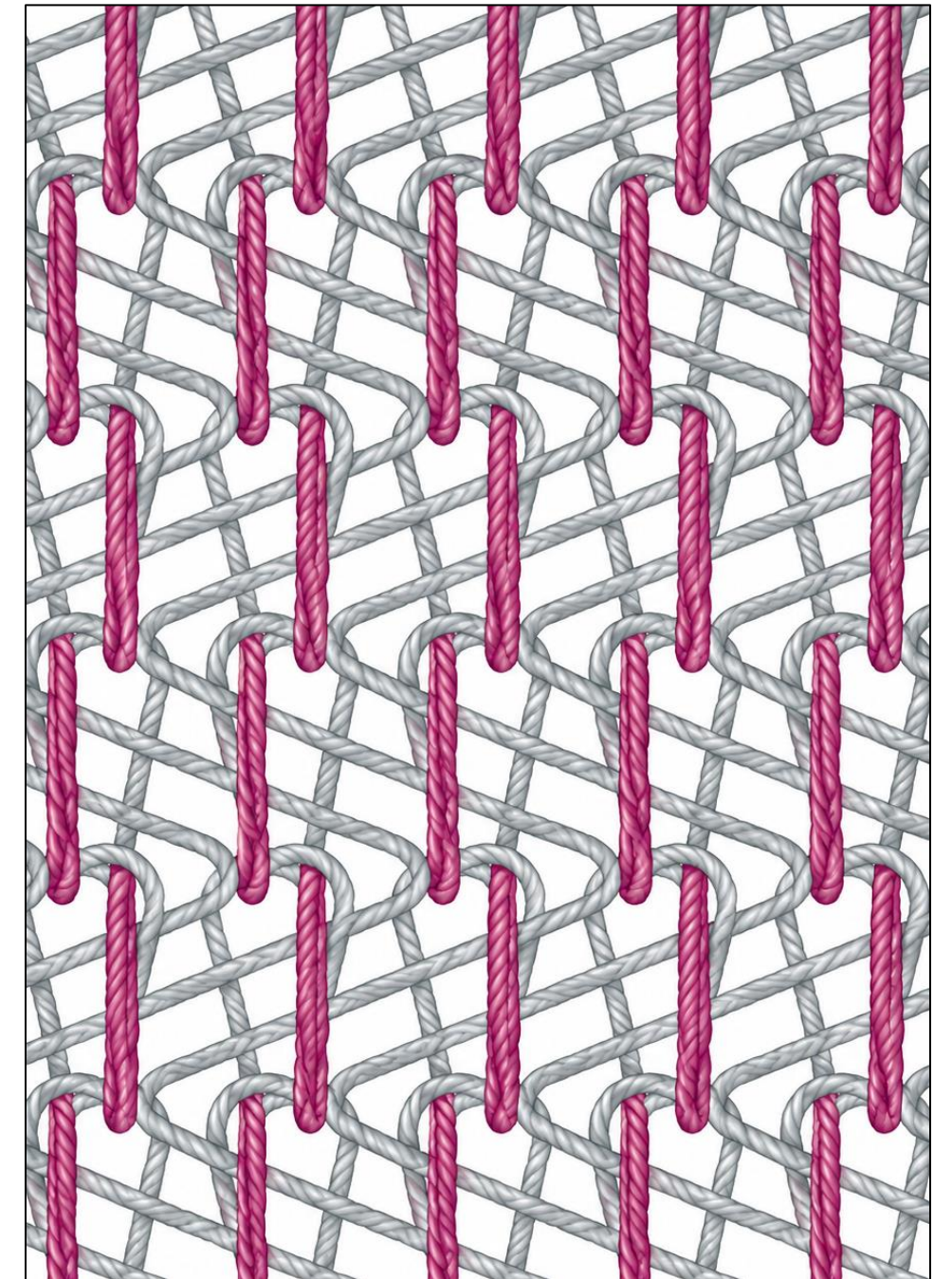
Lapping movement

Front Guide Bar: *Pillar Stitch*.

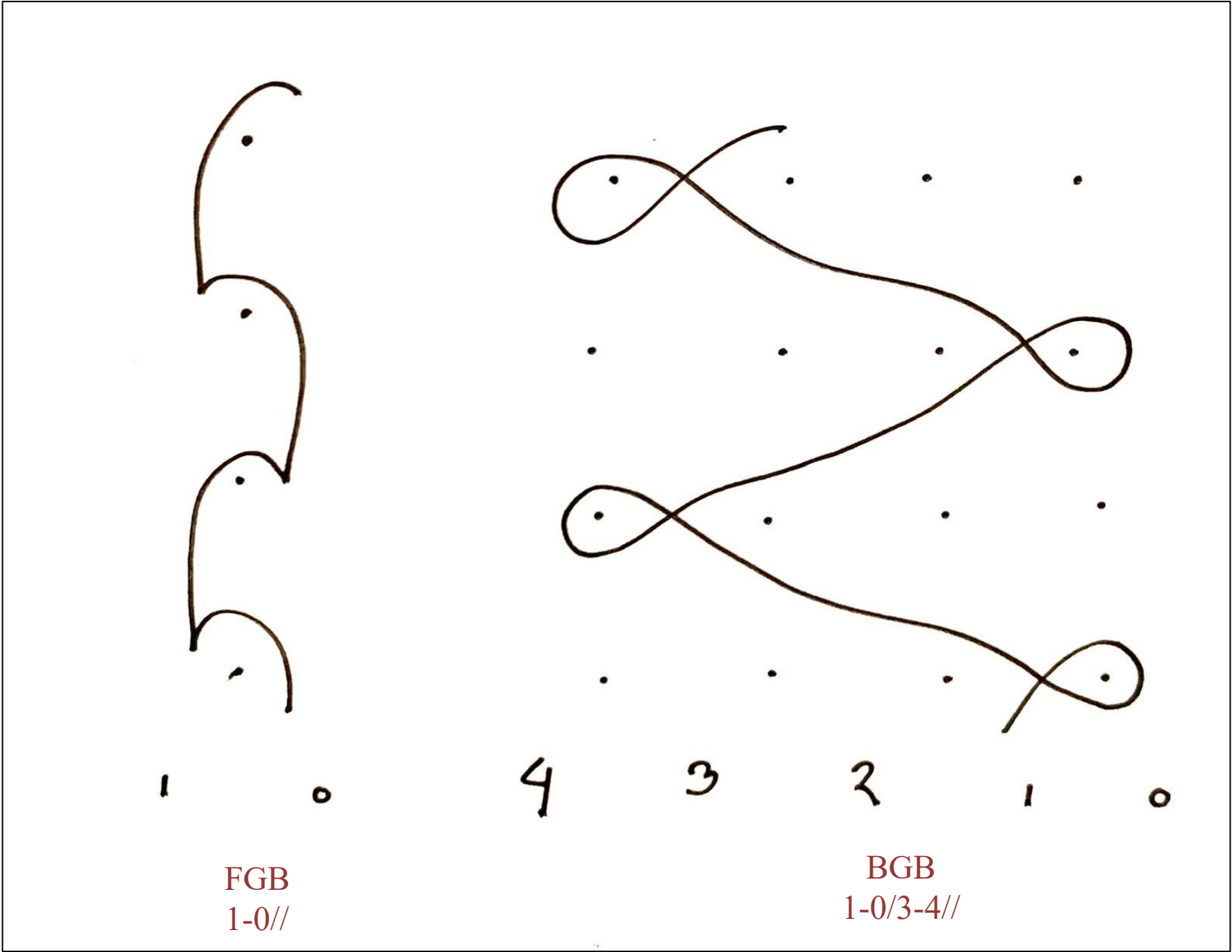
Back Guide Bar: *3 and 1 lap (3 Needle Structure) or 4 and 1 lap (4 Needle Structure)*.

Fabric characteristics

- By threading the front guide bar by colored yarn a pin-stripes effect is produced.
- Superior stability very popular for producing shirting fabric in past.
- Minimum lateral shrinkage.
- Final width of fabric is closed to the knitted width.



QUEENS CORD STRUCTURE



THANK
YOU !

